Haar Random Commutator Bound Conjecture

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1 Introduction

Haar Random Unitaries

We can sample Haar random unitaries as follows:

- Sample a G-ensemble $n \times n$ matrix A.
- Compute the QR decomposition:

$$A = QR.$$

- Let

$$\Lambda = \begin{pmatrix} \frac{R_{11}}{|R_{11}|} & 0 & \cdots & 0 \\ 0 & \frac{R_{22}}{|R_{22}|} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{R_{nn}}{|R_{nn}|} \end{pmatrix}.$$

- Then $Q\Lambda$ is Haar random.

Note: A is invertible with probability 1.

Conjecture

Let U and V be independent Haar random unitaries, and define

$$J := \frac{1}{4}(U + U^* + V - V^*).$$

There exists a universal constant $\gamma > 0$ such that J satisfies

$$\|B\|_2 \leq \gamma \|JB - BJ\|_2,$$

for all self-adjoint matrices B with $\mathrm{Tr}(B) = 0$.

Moreover, this holds with probability approaching 1 as $n \to \infty$.

2 A Toy Model for Non-Normality and Large Pseudospectrum

2.1 Normal Matrices

Recall that a matrix $A \in M_n(\mathbb{C})$ is called ***normal*** if

$$AA^* = A^*A.$$

By the spectral theorem, a normal matrix can be unitarily diagonalized:

$$A = UDU^*,$$

where U is unitary and D is diagonal. Therefore, the eigenvectors of a normal matrix can be chosen to be orthonormal.

For a normal matrix, the resolvent satisfies

$$\|(zI - A)^{-1}\| = \frac{1}{\operatorname{dist}(z, \sigma(A))},$$

where $\sigma(A)$ denotes the spectrum of A. Hence the ε-pseudospectrum is simply the union of ε-balls around the eigenvalues:

$$\sigma_\varepsilon(A) = \{z \in \mathbb{C} : \operatorname{dist}(z, \sigma(A)) < \varepsilon\}.$$

2.2 Resolvent and Pseudospectrum

For a matrix A, the ***resolvent*** is defined by

$$R_A(z) := (zI - A)^{-1},$$

whenever $zI - A$ is invertible.

The **spectrum** of A is

$$\sigma(A) = \{z \in \mathbb{C} : zI - A \text{ is not invertible}\}.$$

The ε-**pseudospectrum** of A is defined by

$$\sigma_\varepsilon(A) = \left\{ z \in \mathbb{C} : \|(zI - A)^{-1}\| > \frac{1}{\varepsilon} \right\}.$$

Equivalently,

$$\sigma_\varepsilon(A) = \{z \in \mathbb{C} : z \in \sigma(A + E) \text{ for some } E \text{ with } \|E\| < \varepsilon\}.$$

Thus the pseudospectrum measures how sensitive the eigenvalues of A are under small perturbations.

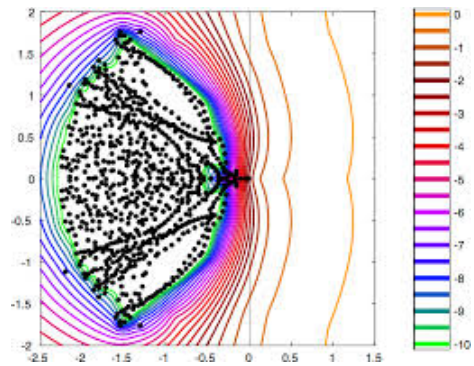


Figure 1: example of psp

2.3 Non-Normal Matrices

If A is diagonalizable but not necessarily normal, then

$$A = VDV^{-1},$$

where V is not necessarily unitary. In this case,

$$zI - A = V(zI - D)V^{-1}.$$

Therefore,

$$(zI - A)^{-1} = V(zI - D)^{-1}V^{-1}.$$

Taking norms gives

$$\|(zI - A)^{-1}\| \leq \|V\|\|V^{-1}\| \cdot \|(zI - D)^{-1}\|.$$

Since

$$\|(zI - D)^{-1}\| = \frac{1}{\operatorname{dist}(z, \sigma(A))},$$

we obtain

$$\|(zI - A)^{-1}\| \leq \kappa(V)\frac{1}{\operatorname{dist}(z, \sigma(A))},$$

where

$$\kappa(V) := \|V\|\|V^{-1}\|$$

is the condition number of the eigenvector matrix.

Hence, if the eigenvectors are highly non-orthogonal, then V is badly conditioned and the resolvent norm may become very large even when z is not extremely close to the eigenvalues. This is the mechanism behind large pseudospectra of non-normal matrices.

2.4 A Simple Construction

Consider the matrix

$$A_M = \begin{pmatrix} 0 & M \\ 0 & 1 \end{pmatrix}, \qquad M \gg 1.$$

This matrix is upper triangular, so its eigenvalues are simply the diagonal entries:

$$\sigma(A_M) = \{0, 1\}.$$

For the eigenvalue $\lambda = 0$, we solve

$$A_M v = 0.$$

Writing

$$v = \begin{pmatrix} x \\ y \end{pmatrix},$$

we get

$$\begin{pmatrix} 0 & M \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} My \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Hence $y = 0$, and we may choose

$$v_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

For the eigenvalue $\lambda = 1$, we solve

$$(A_M - I)v = 0.$$

Since

$$A_M - I = \begin{pmatrix} -1 & M \\ 0 & 0 \end{pmatrix},$$

we need

$$-x + My = 0.$$

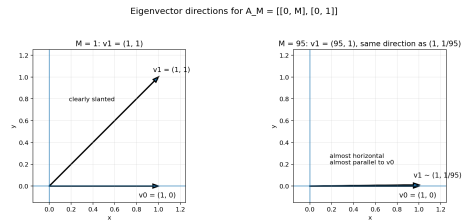
Thus $x = My$, and we may choose

$$v_1 = \begin{pmatrix} M \\ 1 \end{pmatrix}.$$

When M is large,

$$v_1 = \begin{pmatrix} M \\ 1 \end{pmatrix} \approx \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

after normalization. Hence the two eigenvectors become almost parallel. Therefore, even though A_M has only two eigenvalues, its eigenvectors are highly nonorthogonal.



Equivalently, we can write

$$A_M = V \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} V^{-1},$$

where

$$V = \begin{pmatrix} 1 & M \\ 0 & 1 \end{pmatrix}.$$

The columns of V are precisely the eigenvectors

$$v_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad v_1 = \begin{pmatrix} M \\ 1 \end{pmatrix}.$$

As $M \to \infty$, the eigenbasis becomes badly conditioned.

2.5 Resolvent of the Toy Model

We compute

$$zI - A_M = \begin{pmatrix} z & -M \\ 0 & z-1 \end{pmatrix}.$$

Therefore,

$$(zI - A_M)^{-1} = \begin{pmatrix} \frac{1}{z} & \frac{M}{z(z-1)} \\ 0 & \frac{1}{z-1} \end{pmatrix},$$

whenever $z \neq 0, 1$.

The key term is

$$\frac{M}{z(z-1)}.$$

When M is large, this term can be large even if z is not very close to either eigenvalue 0 or 1. Therefore the resolvent norm can be much larger than in the normal case.

Thus, the pseudospectrum of A_M is much larger than just small ε-balls around the eigenvalues.

2.6 Interpretation

This example illustrates the key idea:

$$\text{few eigenvalues} \quad \not\Rightarrow \quad \text{simple spectral behavior.}$$

For normal matrices, the eigenvectors are orthogonal, so the pseudospectrum is controlled by the eigenvalues. For non-normal matrices, however, the geometry of eigenvectors matters. If the eigenvectors are highly non-orthogonal, then the resolvent can be strongly amplified, and the pseudospectrum becomes much larger.

This toy model is not necessarily a malnormal matrix. Rather, it should be viewed as a simple example showing how non-normality and non-orthogonal eigenvectors can lead to large pseudospectrum.

2.7 Resolvent, Pseudospectrum, and the Commutator Inequality

We now clarify the relationship between the resolvent viewpoint and our commutator inequality.

Recall that our matrix is

$$J = \frac{1}{4}(U + U^* + V - V^*).$$

The conjectural inequality says that there exists a universal constant $\gamma > 0$ such that

$$\|B\|_2 \leq \gamma \|JB - BJ\|_2$$

for all self-adjoint matrices B satisfying

$$\mathrm{Tr}(B) = 0.$$

Equivalently, if we define the commutator map

$$\mathrm{ad}_J(B) := JB - BJ,$$

then the conjecture says that ad_J is bounded below on the space of trace-zero self-adjoint matrices. That is, there exists $\kappa > 0$ such that

$$\|\,\mathrm{ad}_J(B)\|_2 \geq \kappa \|B\|_2, \qquad \kappa = \frac{1}{\gamma}.$$

Thus the real obstruction to the conjecture is not the existence of a particular eigenvalue of J. Rather, the obstruction would be the existence of a nonzero trace-zero self-adjoint matrix B which almost commutes with J:

$$JB \approx BJ.$$

More precisely, the conjecture would fail if there exists a sequence (B_n) such that

$$B_n = B_n^*, \qquad \mathrm{Tr}(B_n) = 0, \qquad \|B_n\|_2 = 1,$$

but

$$\|JB_n - B_nJ\|_2 \to 0.$$

Such a sequence would represent an almost-kernel direction for the commutator map

$$B \mapsto [J, B].$$

This also explains why normal matrices cannot satisfy this type of inequality. If J is normal, then it is unitarily diagonalizable:

$$J = UDU^*.$$

In the diagonal basis, many non-scalar diagonal matrices commute with D. For example, if

$$D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

then

$$DB = BD.$$

After conjugating back by U, we obtain a nonzero trace-zero self-adjoint matrix which commutes with J. Hence

$$[J, B] = 0$$

while

$$\|B\|_2 > 0.$$

Therefore the inequality

$$\|B\|_2 \leq \gamma \|[J, B]\|_2$$

would become

$$\|B\|_2 \leq 0,$$

which is impossible.

Thus the conjecture requires J to be strongly non-normal. More precisely, it requires that J have no non-scalar trace-zero self-adjoint matrices that commute, or even almost commute, with it.

On the other hand, the resolvent of J is

$$R_J(z) = (zI - J)^{-1}.$$

The resolvent studies the behavior of the operator

$$zI - J.$$

When

$$\|(zI - J)^{-1}\|$$

is large, the matrix $zI - J$ is close to being singular. This means that z behaves like an approximate eigenvalue. This is the idea behind the ε-pseudospectrum:

$$\sigma_\varepsilon(J) = \left\{ z \in \mathbb{C} : \|(zI - J)^{-1}\| > \frac{1}{\varepsilon} \right\} \cup \sigma(J).$$

However, the resolvent is not the same object as the commutator map. The resolvent studies

$$zI - J : \mathbb{C}^n \to \mathbb{C}^n,$$

whereas our conjecture studies

$$\mathrm{ad}_J : B \mapsto JB - BJ$$

on the space of trace-zero self-adjoint matrices.

Therefore, a large pseudospectrum does not directly prove the commutator inequality, nor does it directly disprove it. Instead, it provides evidence of non-normal behavior. Large pseudospectra indicate that the matrix is highly non-normal, while the conjecture requires an even stronger property: that there are no trace-zero self-adjoint almost-commuting directions.

In summary, the resolvent detects instability of

$$zI - J,$$

while the commutator inequality detects instability of

$$B \mapsto [J, B].$$

These are related through non-normality, but they are not the same object.

The logical chain can be summarized as follows:

$$\text{normality} \to \text{many commuting self-adjoint matrices} \to \exists B \neq 0,\ \mathrm{Tr}(B) = 0,\ [J, B] = 0$$

$$\longrightarrow \text{the commutator inequality fails.}$$

Therefore, to prove the conjecture, one needs to show

no nonzero trace-zero self-adjoint B can almost commute with J.

Equivalently,

$$\mathrm{ad}_J : B \mapsto [J, B]$$

must have no almost-kernel on the trace-zero self-adjoint subspace.